CCNA Networking Internship – Task 1

Build Your CCNA Lab + How Networks Actually Work

Organization: TechBiz
Week: 1
Task: 1
Topic: Network Fundamentals and Basic Network Configuration
Software: Cisco Packet Tracer

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1. Introduction

The objective of this lab was to build a basic network topology using Cisco Packet Tracer, configure IPv4 addressing, establish communication between different networks, and understand how network devices communicate with one another.

The lab consists of two routers, two switches, and four PCs. The two LANs are connected through routers, allowing devices on different networks to communicate using static routing.

2. Objectives

The main objectives of this lab were:

- Install and explore Cisco Packet Tracer.
- Build a network topology consisting of two routers, two switches, and four PCs.
- Configure IPv4 addresses and subnet masks for all end devices and network interfaces.
- Establish communication between two different LANs.
- Configure static routing between the routers.
- Test connectivity using ICMP ping commands.
- Understand the functions of network devices and the journey of a packet across a network.

3. Network Topology

The topology was designed using Cisco Packet Tracer with two separate LANs connected through two routers.

Devices used:

Device	Model	Quantity
Routers	Cisco 2911	2
Switches	Cisco 2960	2
PCs	Generic PC	4

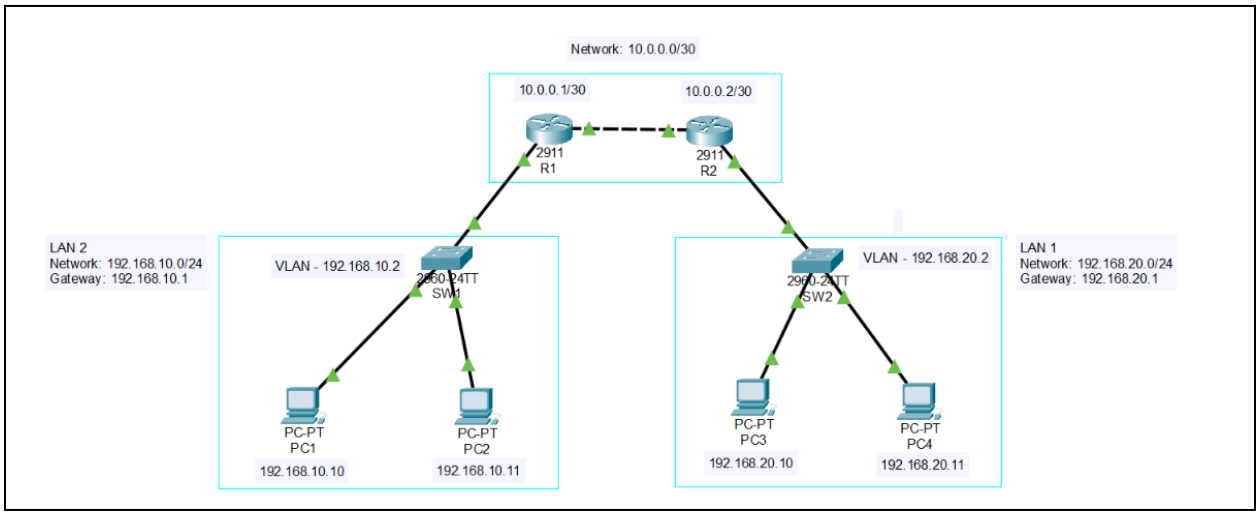


Figure 1: Complete network topology created in Cisco Packet Tracer.

The network was divided into three IP networks:

- LAN 1: 192.168.10.0/24
- Router-to-router link: 10.0.0.0/30
- LAN 2: 192.168.20.0/24

PC1 and PC2 are connected to SW1, which connects to R1. Similarly, PC3 and PC4 are connected to SW2, which connects to R2. The two routers communicate through a direct Ethernet connection.

4. Device Functions

4.1 Router

A router connects different networks and forwards packets between them. It examines destination IP addresses and uses its routing table to determine the appropriate path for forwarding traffic.

In this lab, R1 and R2 connect the two LANs and enable communication between them using static routes.

4.2 Switch

A switch connects multiple devices within a LAN. It forwards Ethernet frames based on MAC addresses and helps devices within the same local network communicate.

SW1 connects PC1 and PC2, while SW2 connects PC3 and PC4.

4.3 Personal Computer (PC)

A PC is an end device that sends and receives data over a network. Each PC was assigned a unique IPv4 address, subnet mask, and default gateway to enable network communication.

4.4 Hub (Additional Concept)

A hub is a basic networking device that connects multiple devices but does not intelligently filter traffic. It broadcasts incoming data to all connected ports, unlike a switch, which forwards frames based on MAC addresses.

5. IP Addressing Configuration

The following IPv4 addressing scheme was used throughout the topology.

5.1 Router Addressing

Router	Interface	IP Address	Subnet Mask
R1	G0/0	192.168.10.1	255.255.255.0
R1	G0/1	10.0.0.1	255.255.255.252
R2	G0/0	192.168.20.1	255.255.255.0
R2	G0/1	10.0.0.2	255.255.255.252

5.2 PC Addressing

PC	IPv4 Address	Subnet Mask	Default Gateway
PC1	192.168.10.10	255.255.255.0	192.168.10.1
PC2	192.168.10.11	255.255.255.0	192.168.10.1
PC3	192.168.20.10	255.255.255.0	192.168.20.1
PC4	192.168.20.11	255.255.255.0	192.168.20.1

5.3 Switch Management Addressing

Switch	Interface	IP Address	Subnet Mask	Default Gateway
SW1	VLAN 1	192.168.10.2	255.255.255.0	192.168.10.1
SW2	VLAN 1	192.168.20.2	255.255.255.0	192.168.20.1

The /24 subnet masks were used for the two LANs, while a /30 subnet mask was used for the point-to-point router connection.

6. Device Configuration

6.1 Router Configuration

Both routers were configured through the Cisco IOS Command Line Interface (CLI).

The following commands were used on R1:

```
None
enable
configure terminal
hostname R1

interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
```

R2 was configured similarly, using its assigned IP addresses.

The no shutdown command was used to enable the router interfaces.

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.20.1    YES manual up                    up
GigabitEthernet0/1     10.0.0.2        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Figure 2: Router interface configuration verified using the show ip interface brief command.

6.2 Switch Configuration

The switches were assigned management IP addresses through their VLAN 1 interfaces.

For SW1:

```
None
enable
configure terminal
hostname SW1

interface vlan 1
ip address 192.168.10.2 255.255.255.0
no shutdown
exit

ip default-gateway 192.168.10.1
```

SW2 was configured similarly with its corresponding IP address and default gateway.

6.3 PC Configuration

The IPv4 addresses, subnet masks, and default gateways were configured manually through the Desktop > IP Configuration section of each PC in Packet Tracer.

The addressing information was assigned according to the table in Section 5.

7. Static Routing Configuration

Initially, the two LANs were separate networks. To enable communication between them, static routes were configured on both routers.

On R1:

```
ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

This route directs traffic destined for LAN 2 through R2.

On R2:

```
ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

This route directs traffic destined for LAN 1 through R1.

The routing tables were verified using:

```
show ip route
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
S    192.168.20.0/24 [1/0] via 10.0.0.2
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
S    192.168.10.0/24 [1/0] via 10.0.0.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0
```

Figure 3: Static routes displayed in the routing tables of R1 and R2.

8. Connectivity Testing

Connectivity between the devices was tested using the ICMP ping command.

The following tests were performed:

Source	Destination	Destination IP	Result
PC1	PC2	192.168.10.11	Successful
PC1	PC3	192.168.20.10	Successful
PC1	PC4	192.168.20.11	Successful
PC2	PC3	192.168.20.10	Successful
PC2	PC4	192.168.20.11	Successful
PC3	PC4	192.168.20.11	Successful

The ping command was used to verify connectivity between devices.

Example command from PC1:

```
None
ping 192.168.20.11
```

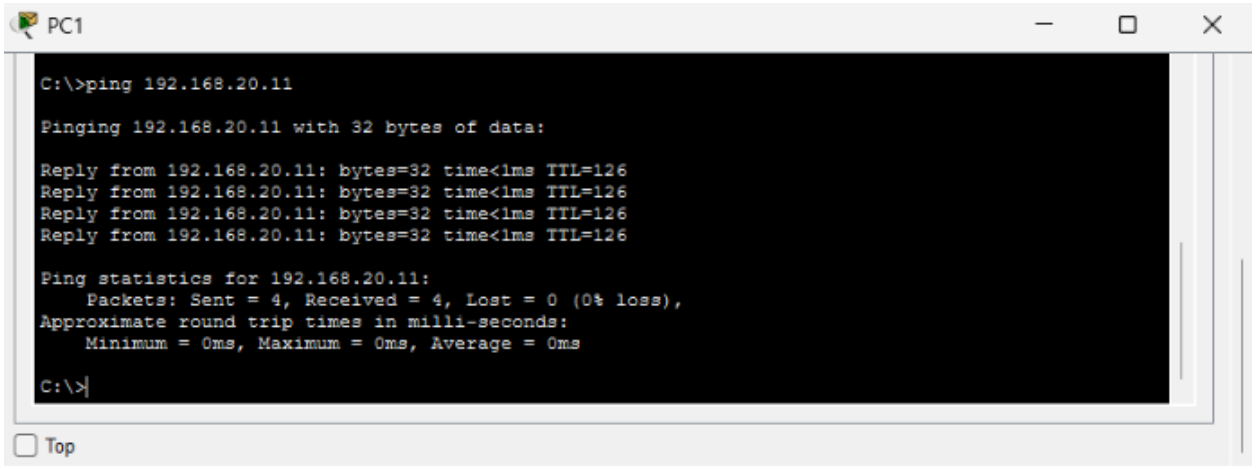


Figure 4: Successful ICMP ping from PC1 to PC4 across the two LANs.

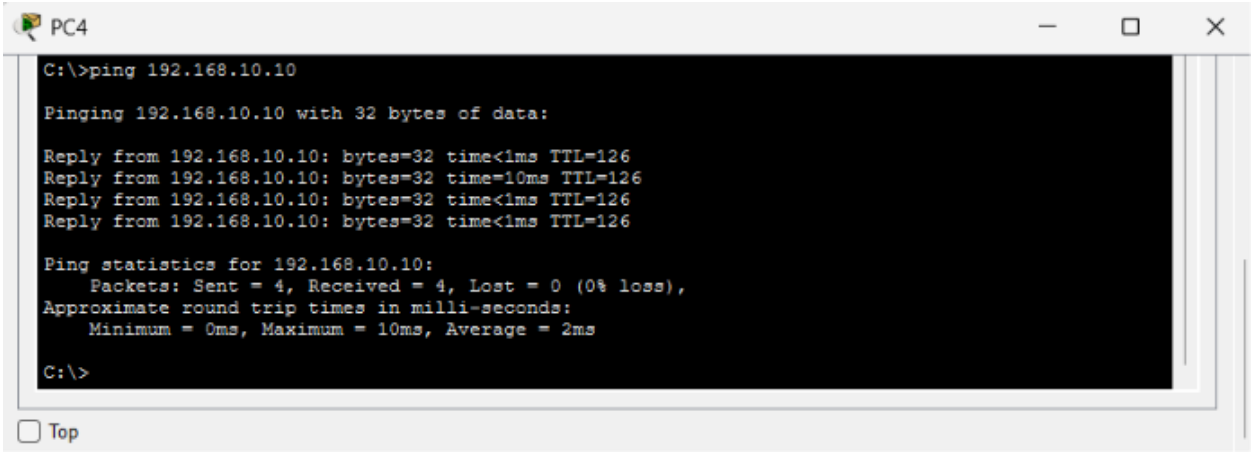


Figure 5: Successful connectivity test from PC4 to PC1.

All four PCs were able to communicate successfully, confirming that the IP addressing and static routing configurations were working as intended.

9. Packet Journey: PC1 to PC4

When PC1 sends a ping request to PC4, the packet travels through the network as follows:

1. **PC1:** Generates an ICMP Echo Request addressed to PC4 (192.168.20.11). Since PC4 belongs to a different network, PC1 forwards the packet to its default gateway, R1 (192.168.10.1).
2. **SW1:** Receives the Ethernet frame from PC1 and forwards it towards R1 based on its MAC address table.
3. **R1:** Receives the packet, checks its routing table, and identifies the static route to the 192.168.20.0/24 network. It forwards the packet to R2 through the 10.0.0.0/30 network.
4. **R2:** Receives the packet and forwards it towards LAN 2 through its G0/0 interface.
5. **SW2:** Receives the Ethernet frame and forwards it to PC4.
6. **PC4:** Receives the ICMP Echo Request and generates an ICMP Echo Reply.

The reply travels back through SW2, R2, R1, SW1, and finally reaches PC1.

Packet Path:

PC1 → SW1 → R1 → R2 → SW2 → PC4

This demonstrates how routers forward packets between different networks, while switches forward Ethernet frames within their respective LANs.

10. Conclusion

This lab provided practical experience in building and configuring a basic network using Cisco Packet Tracer.

I successfully created a topology consisting of two routers, two switches, and four PCs. I configured IPv4 addressing, enabled router interfaces, assigned management IP addresses to switches, and established communication between two separate LANs using static routing.

Connectivity was verified through successful ping tests between all four PCs.

Through this exercise, I gained a better understanding of network devices, IPv4 addressing, default gateways, static routing, and how packets travel between different networks.

End of Report